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Report-Supervised Distillation for Multi-Label Knee MRI Classification: An Evaluation Under Limited Expert Annotation

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ABSTRACT Expert image-level labels are scarce for multi-label knee MRI, whereas radiology reports are abundant. We ask how report-consensus targets, self-distillation and distillation from an external image-input model affect a study-level classifier when only 58 of 4,407 training studies carry expert labels for 12 findings (RSNA Knee Abnormality dataset). A publicly available 2.5D ConvNeXt reader was trained on (i) the mean of four language-model report labellers, (ii) a mixture of these labels with its own out-of-fold predictions, and (iii) mixtures with a public CoAtNet teacher whose training used report labels and, for three findings, expert MRI grades from an external cohort. On a 385-study development partition of the hidden test set, which also informed model development, the three targets gave macro ROC AUCs of 0.933, 0.939 and 0.948–0.949; the teacher displayed 0.949. Changes of backbone, pre-training and seed showed no clear benefit in the configurations evaluated. Rank fusion of the students with public ensembles ranked above the unchanged parent in eight of eight development comparisons. On the held-out 70% final-test partition, matched parents with and without students scored [TODO: x.xxx–x.xxx vs. x.xxx–x.xxx], and the readers A, B and C3 scored [TODO:]. Bootstrap intervals on the 58 expert-labelled studies were about 0.05 wide, and a simulation illustrates that a partition of this size can mis-order models whose AUCs differ by about 0.001. Development comparisons are therefore exploratory. Code and experiment logs are released.

INDEX TERMS Knee MRI, knowledge distillation, multi-label classification, pseudo-labelling, radiology reports, weak supervision, model evaluation.

I. INTRODUCTION

KNEE MRI is the reference non-invasive examination for internal derangement of the knee, and a single study is expected to address ligaments, menisci, cartilage in three compartments, joint fluid, the synovium, popliteal cysts and bone marrow. Deep learning has matched radiologists on single tasks such as anterior cruciate ligament (ACL) and meniscal tears and cartilage lesions [1]–[5], and multi-finding models have been trained on expert-annotated cohorts [6], [7]. These systems depend on image-level labels from expert readers, which are costly to obtain at scale.

Radiology reports are abundant but are an imperfect proxy: radiologists omit findings they consider irrelevant, hedge, and grade severity in free text. The RSNA Knee Abnormality dataset [8] makes the gap explicit. Of its 4,407 training studies, all have a report, and 58 additionally have expert labels for 12 findings, which the organisers stated were produced by an image-based process independent of the reports. A model trained on report labels is therefore optimised for a target that differs from the one used to evaluate it.

This paper addresses one question: *under limited expert annotation, how do report consensus, self-distillation and external image-model distillation affect knee MRI classification?* Our hypothesis was that the training target, rather than the backbone, limits a report-supervised reader. The contributions are:

1) A controlled comparison of training targets for a fixed 2.5D reader—report consensus, out-of-fold self-distillation, and an external image-input teacher at two mixing weights—with every change logged prospec-

tively as a single variable against a scored parent (Sections IV-E and VI-B).
2) An evaluation of whether such students add information to strong existing ensembles, on the development partition and on the held-out final-test partition (Sections VI-A and VI-D).
3) An analysis of what the available evaluation data can resolve, with negative results on models whose out-of-fold performance did not translate into ensemble benefit (Sections VI-E and VI-F).

What was reused

The reader architecture, its preprocessing and its training code were released publicly by another author [9]; the report label tables [10], the teacher models [11], [12] and the ensembles used for fusion [12], [13] are likewise public. Our work is the experimental design, the target and teacher comparisons, the fusion experiments, the analyses, and implementation changes needed to run them (Fig. 1).

II. RELATED WORK

A. DEEP LEARNING FOR KNEE MRI

MRNet [1] applied per-plane 2D CNNs with slice pooling to abnormality, ACL-tear and meniscal-tear detection; ELNet [14] proposed a lighter slice-wise network. Task-specific models address cartilage lesions [2], meniscal and patellofemoral degeneration [3], ACL severity [4] and meniscal tears with a surgical reference [5], [15]. Astuto et al. [6] graded several tissues in one pipeline, and Qiu et al. [7] diagnosed twelve knee abnormalities with co-plane attention across sequences (overall AUC 0.812 on 1,748 subjects). External cohorts—the Osteoarthritis Initiative (OAI) [16], fastMRI [17], SKM-TEA [18]—and radiology pre-training [19] are widely used. All of these works rely on image-level labels for the target task, which our setting largely lacks.

B. LEARNING FROM RADIOLOGY REPORTS

Rule-based [20], [21] and transformer [22] report labellers produced the first large chest radiograph datasets, and LLMs can structure free-text reports [23]. For spinal MRI, image classifiers trained on LLM-extracted report labels performed comparably to classifiers trained on clinician annotations [24]. Report labels inherit omissions and hedging, which CheXpert handled with an uncertainty class [21].

C. PSEUDO-LABELLING AND DISTILLATION

Knowledge distillation trains a student on a teacher's soft outputs [25]; born-again networks [26] and Noisy Student [27] showed that students of similar capacity can match or exceed their teachers. Pseudo-labelling [28] and bootstrapped targets that mix noisy labels with predictions [29] are close relatives; our targets are of the latter form.

D. CLOSEST WORK ON THIS DATASET

Kooklan [30] analysed the same dataset and compared report-derived targets with the out-of-fold predictions of a public image model against the 58 expert-labelled studies. The report labels and the image model reached similar agreement with the experts (macro AUC 0.892 and 0.896) but failed on different findings, and their equal mixture agreed better (0.927). However, a 2.5D ResNet-50 trained on the mixture did *not* outperform the same model trained on report labels on the final-test partition (0.872 vs 0.877). Our study differs in the reader (a 2.5D multi-slot ConvNeXt transformer), in the teachers (an in-sample CoAtNet teacher with external expert grades, and a self-teacher), in the evaluation of fusion with ensembles, and in its development-partition result, which points the other way. Reconciling the two findings requires final-test scores of our students (Section VI-A).

E. EVALUATION WITH SMALL OR HIDDEN TEST SETS

Repeated evaluation on a fixed partition invites adaptive over-fitting [31], although its extent is often smaller than feared [32]. Rankings in biomedical challenges are sensitive to test sampling and metric choice [33], and small test sets are a recurring methodological failure [34]. We report the reporting items of the CLAIM checklist [35] in the Supplementary Material.

III. DATA

A. COHORT, REFERENCE STANDARD AND LABEL DEFINITIONS

The RSNA Knee Abnormality dataset [8] is an international, multilingual collection of knee MRI examinations, each paired with its original radiology report [36]. It contains 4,407 de-identified training studies (24,371 DICOM series) and a hidden test set of about 1,284 studies. Twelve binary findings are scored per study under the organisers' names: ACL, MCL, medial meniscus, lateral meniscus, medial OA, lateral OA, patellofemoral (PF) OA, effusion, synovitis, Baker's cyst, contusion and fracture. Each series carries its plane and a fat-suppression (FS) flag; Fig. 2 illustrates the planes and sequence types with openly licensed images, because the dataset's terms forbid publishing its images.

Reference standard

According to the organisers [36], every study in the reference set was labelled independently by two subspecialty-trained musculoskeletal radiologists against a written annotation rubric, with disagreements adjudicated by a third radiologist; labels apply to the whole examination of a single knee. The organisers stated separately that the same process was used for the test set [37]. Borderline findings were graded negative to favour specificity. Table 1 summarises the rubric. Several definitions are narrower than the clinical terms: the "ACL" and "MCL" targets are high-grade tears, "OA" requires high-grade cartilage loss over about 1 cm or more, and effusion and Baker's cyst must be at least moderate. Reports

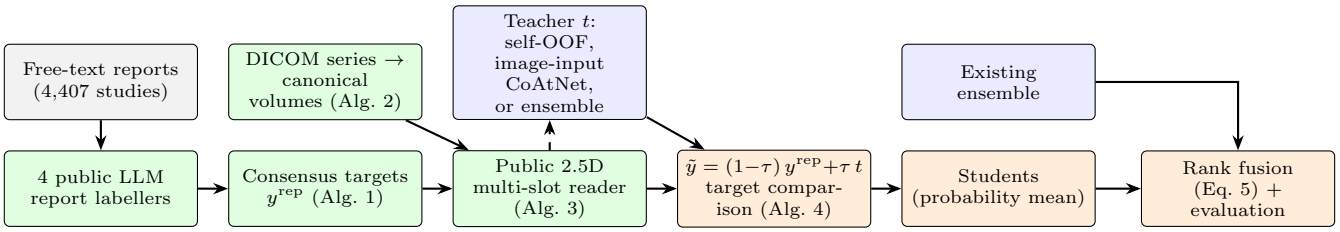


FIGURE 1. **Overview. Green: reused public components (report labels, preprocessing and reader architecture); blue: external teachers and ensembles; orange: components of this work (target construction and comparison, fusion, evaluation). The dashed arrow denotes self-distillation, in which the teacher is the reader's own out-of-fold prediction.**

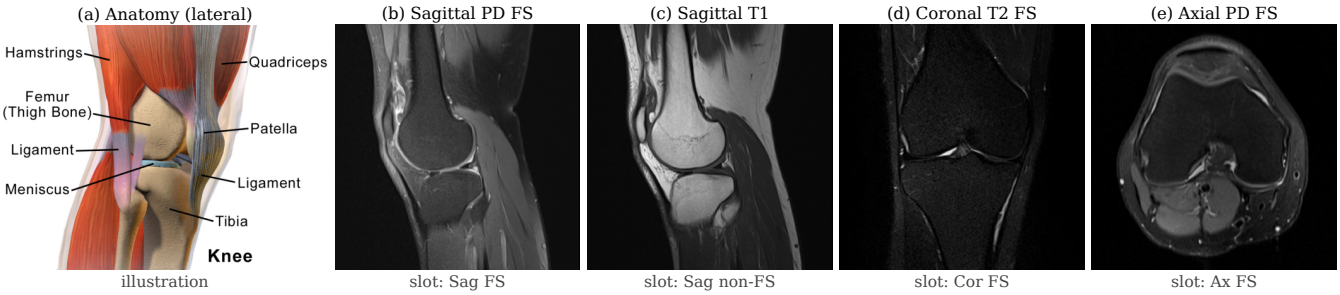


FIGURE 2. **Illustration of knee anatomy and of the MRI planes and sequences that map to the reader's slots (Section IV-C). *These images are not from the RSNA dataset*, whose terms forbid publishing it. (a) Blausen.com staff, "Medical gallery of Blausen Medical 2014," *WikiJournal of Medicine*, 2014, doi:10.15347/wjm/2014.010, CC BY 3.0. (b)–(e) Knee MRI, Siemens MAGNETOM Aera 1.5 T, by Ptrump16 via Wikimedia Commons, CC BY-SA 4.0; (d) and (e) are middle frames of the published coronal and axial series. Cropped and arranged; this figure is licensed CC BY-SA 4.0.**

TABLE 1. **Positive definitions of the reference labels, summarised from the organisers' annotation rubric [36]. Borderline findings are negative.**

Finding	Positive if
ACL	high-grade partial (>50% of fibres) or complete tear; signal change or thickening alone is negative
MCL	high-grade partial or complete acute tear with fibre disruption and edema; low-grade sprain or chronic change is negative
Medial / lateral meniscus	signal definitely reaching the meniscal surface on $\geq$2 images, or a truncated, diminutive or displaced fragment; intrasubstance degeneration is negative
Medial / lateral / PF OA	moderate or large area ($\gtrsim$1 cm) of cartilage loss >50% of thickness in that compartment
Effusion	moderate or large joint fluid distending the joint
Synovitis	inflammation and thickening of the synovial lining
Baker's cyst	moderate or large fluid collection in the popliteal location
Contusion	bone-marrow edema-like signal from impact without a fracture line
Fracture	acute cortical break or fracture line

describe milder grades, which is one source of the report–reference gap (Section III-D).

What is not reported

The materials available to us do *not* report the contributing institutions, acquisition period, demographics, scanner vendors and field strengths, inclusion and exclusion criteria, patient identifiers, repeated or bilateral examinations, or how the 58 reference studies were chosen. We mark these items as unavailable in the CLAIM checklist rather than infer them.

TABLE 2. **Positive studies per finding in the 58 expert-labelled studies ($\mathcal{G}_{58}$).**

Finding	n_+	Finding	n_+	Finding	n_+
ACL	24	Medial OA	15	Synovitis	27
MCL	9	Lateral OA	11	Baker's cyst	12
Med. meniscus	26	PF OA	21	Contusion	19
Lat. meniscus	23	Effusion	35	Fracture	18

Grouping studies with identical reports (Section IV-A) removes duplicated reports but does not establish patient-level separation.

B. THE 58 EXPERT-LABELLED STUDIES

Expert labels are available for 58 training studies ($\mathcal{G}_{58}$). The set is enriched: every study has at least one positive finding, with 4.14 on average (Table 2). $\mathcal{G}_{58}$ was never used for gradient updates of any model trained in this work. It was, however, used during development: (i) as a diagnostic reported for every model; (ii) as part of some adoption rules, for example "adopt a student only if its $\mathcal{G}_{58}$ AUC is at least 0.9127" or "reject a change if $\mathcal{G}_{58}$ falls by more than 0.01"; and (iii) by the authors of the external teacher, who used it for model selection [11]. Its scores and intervals are therefore descriptive development results, not independent estimates.

C. REPORT-DERIVED LABELS

Several groups released LLM-extracted labels for all training studies [10]; for example, the number of fracture-positive

TABLE 3. Agreement between two report-derived label sets on 4,349 training studies (inter-source agreement, not accuracy): AUC of set D against set T binarised at 0.5, Cohen's κ at 0.5, and positive rates ($\geq$ 0.5) of each set.

Finding	AUC	κ	D pos.	T pos.
ACL	0.950	0.70	0.13	0.21
MCL	0.983	0.44	0.05	0.15
Medial meniscus	0.966	0.84	0.38	0.42
Lateral meniscus	0.967	0.80	0.14	0.16
Medial OA	0.979	0.69	0.25	0.37
Lateral OA	0.969	0.61	0.15	0.27
PF OA	0.987	0.73	0.34	0.46
Effusion	0.919	0.38	0.26	0.60
Synovitis	**0.619**	0.31	0.25	0.13
Baker's cyst	0.986	0.59	0.12	0.25
Contusion	0.891	0.55	0.15	0.17
Fracture	0.979	0.68	0.04	0.07
Macro	0.933	0.61		

studies ranges from 159 to 336 across five tables. Table 3 compares two label sets on the 4,349 studies outside $\mathcal{G}_{58}$: *set D*, the soft labels `labels_llm_soft` of dreaddevelopment [38], used by our EfficientNetV2-S models (Section VI-E); and *set T*, a team consensus of four public tables (Section III-D). Neither is the reader's target v0 (Section IV-A). The table measures agreement between label sources, not accuracy against expert labels. The two sets order studies similarly for 11 findings (AUC 0.89–0.99) but not for synovitis (0.62), because one imputes synovitis from effusion when the report is silent. For effusion the disagreement is a threshold: "small effusion" is about 0.2 in one set and 0.85 in the other, which flips 34% of studies at 0.5 while preserving their order.

D. EXPLORATORY ANALYSIS OF THE LABELS

We analysed the labels that were available locally (no images). The source tables of the reader's target v0 are not stored in our repository, so Figs. 3–5 use a *different* construction: set T, a team consensus of four public tables (mri-balanced-labels-public, jev-knee-labels `HYBRID_labels_scores`, hybrid-report-labels `report_labels_v4hybrid` and stevenleehans `llm_labels_v4_blend`), which were accepted because their $\mathcal{G}_{58}$ macro AUC was within 0.03 of the best table; one post-processing rule (a cap on medial OA when the report is silent) was also kept because it raised $\mathcal{G}_{58}$ AUC [39]. Set T shares one source with v0 and was used for training only by the E03b model. These figures therefore describe report-derived labels in this cohort, not the reader's training targets, and set T's agreement with $\mathcal{G}_{58}$ is not reported because $\mathcal{G}_{58}$ shaped it. Fig. 6 uses a benchmark of public tables against $\mathcal{G}_{58}$ computed by the same pipeline.

Label composition

Fig. 3 shows, for each finding, the share of studies whose set-T label is negative (< 0.25), uncertain (0.25–0.7) or positive ($\geq$ 0.7). Positive shares range from 5% (fracture) to 58% (effusion). Synovitis is the most ambiguous finding: 49% of

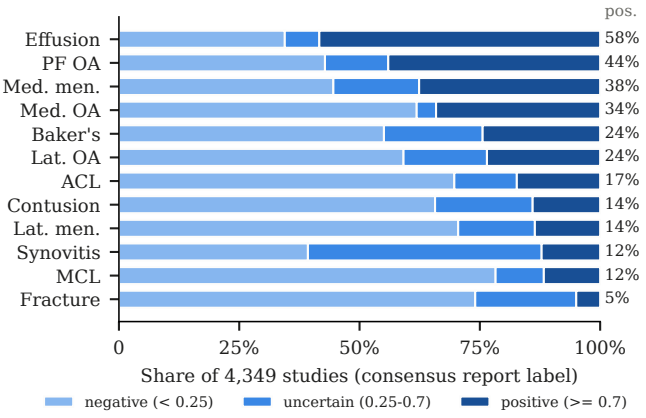


FIGURE 3. Composition of the team-consensus labels (set T; not the reader's targets v0) of the 4,349 report-labelled studies, by finding. Right: positive share.

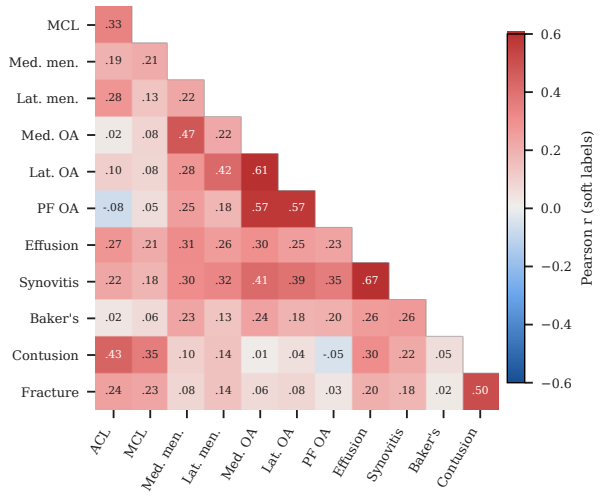


FIGURE 4. Pearson correlation between the team-consensus soft labels (set T) of the 12 findings (4,349 studies).

studies fall in the uncertain band, compared with 4–21% for the other findings, in line with its low inter-source agreement (Table 3).

Co-occurrence

Findings co-occur in clinically expected patterns (Fig. 4): effusion and synovitis ($r = 0.67$), the three OA compartments (0.57–0.61), contusion and fracture (0.50), and medial meniscus with medial OA (0.47); ACL is most associated with contusion (0.43) and MCL (0.33). These dependencies are one reason why a shared study representation with per-finding attention, rather than 12 independent models, is a natural design.

Findings per study

With a positive threshold of 0.7 on set T, report-labelled studies carry 2.96 positive findings on average and 14% carry none, whereas every $\mathcal{G}_{58}$ study has at least one expert-positive finding (mean 4.14; Fig. 5). The two populations and label

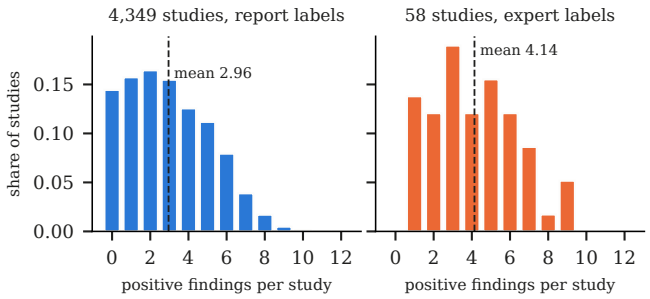


FIGURE 5. Number of positive findings per study: team-consensus labels (set T, ≥ 0.7) of the 4,349 report-labelled studies (left) and expert labels of the 58 reference studies (right). Dashed: mean.

types differ, so the comparison illustrates that $\mathcal{G}_{58}$ is enriched, not that reports under-call findings.

Report labellers versus expert labels

Fig. 6 compares 12 public report-label tables with the expert labels of $\mathcal{G}_{58}$. Three further tables were flagged by the pipeline's heuristic ($\mathcal{G}_{58}$ macro AUC > 0.995, or "gold" in the source name; all three reached AUC 1.0). Perfect ranking does not prove that a table contains the released reference labels, and we did not verify cell-by-cell equality; we treat them as potentially reference-contaminated and conservatively excluded them. Macro AUCs of the remaining tables range from 0.81 to 0.91. Agreement is high for ACL, MCL, medial meniscus and Baker's cyst (median 0.93–0.98) and lowest for synovitis (median 0.71), fracture (0.82), contusion (0.83) and lateral OA (0.83), the findings for which reports are most often silent or hedged. The four tables averaged into the reader's targets (Eq. 1, marked $*$) span 0.82–0.89. With 58 studies, differences of a few hundredths between tables are within sampling noise (Section VI-F).

E. DEVELOPMENT AND FINAL-TEST PARTITIONS

Test labels are not released. A model is submitted as an offline inference program limited to 9 h on two NVIDIA T4 GPUs, and the organisers' server computes the macro ROC AUC [40] (the mean over findings of the per-finding AUC). Scores on a fixed 30% partition (about 385 studies, displayed at three decimals) were visible during development; we call this the *development partition*. More than 60 configurations were scored on it, and its scores informed which changes were kept, so it is not an independent test. Scores on the remaining 70% (about 899 studies) became visible only after all model and final-selection decisions had been made; we call this the *final-test partition*. It is a held-out evaluation not used for model development or final selection; we make no claim about patient overlap with the training set or about the representativeness of the partition, which the organisers do not report. Fig. 7 summarises how each part of the data was used.

When two development scores display the same value, we used the server's unrounded ordering of our own submissions; such orderings are ordinal and exploratory.

Algorithm 1 Report-consensus targets and report-grouped folds

Input: studies $\mathcal{D}$, reports R_i; label tables $\{y^{(s)}\}_{s\in\mathcal{S}}$; reference set $\mathcal{G}_{58}$; $K_f = 5$

Output: targets $y^{\text{rep}} \in [0.01, 0.99]^{N\times 12}$; fold index $\phi(i) \in \{-1, 0, \ldots, 4\}$

1: **for** each study $i \notin \mathcal{G}_{58}$ and finding c **do**
2: $S_{ic} \leftarrow \{s : y_{ic}^{(s)} \text{ defined}\}$; $y_{ic}^{\text{rep}} \leftarrow \text{clip}\left(\text{mean}_{s\in S_{ic}} y_{ic}^{(s)}, 0.01, 0.99\right)$
3: **end for**
4: $g_i \leftarrow \text{Normalise}(R_i)$ ▷ lower-case, strip non-alphanumerics
5: $\phi \leftarrow \text{GroupKFold}(K_f)$ on studies $\notin \mathcal{G}_{58}$, fixed permutation, groups g_i
6: $\phi(i) \leftarrow -1$ for all $i \in \mathcal{G}_{58}$
7: **return** y^{rep}, ϕ

IV. METHODS

Fig. 1 separates the reused components from those developed in this work. Let $\mathcal{D} = \{(X_i, R_i)\}_{i=1}^{N}$ be the training studies with image series X_i and report R_i, and $\mathcal{C}$ the 12 findings. The goal is a scoring function $f : X \mapsto [0,1]^{12}$ whose per-finding rankings maximise the macro AUC on unseen studies.

A. REPORT-CONSENSUS TARGETS

We adopted the consensus construction of the public reader [9], which averages four public LLM label tables [10]: stevenleehans `llm_labels_v4_blend`, pilkwang `report_labels_v2`, lixin73 `labels_llm_gpt56sol` and laymond `qwen_knee_weak_labels`. We call the result *v0*. With $y_{ic}^{(s)} \in [0,1]$ the label of source s for study i and finding c, and S_{ic} the sources that define it,

$$y_{ic}^{\text{rep}} = \text{clip}\left(\tfrac{1}{|S_{ic}|}\textstyle\sum_{s\in S_{ic}} y_{ic}^{(s)},\ 0.01,\ 0.99\right). \qquad (1)$$

We did not compare this consensus with each individual source under the same protocol, so we make no claim about its benefit. Folds group studies with identical normalised report text (Algorithm 1).

B. CANONICAL PREPROCESSING (REUSED)

Each series is mapped to a canonical volume by one function shared by training and inference (Algorithm 2). Slices are ordered by position along the slice normal; each slice is transposed and flipped so that its rows and columns follow the nearest canonical patient (LPS) axes. Oblique acquisitions are therefore re-oriented only to the nearest axis and are not resampled into a common frame. Images are resampled to 0.4 mm pixels and centre-cropped to 153.6 mm (384×384 pixels). Intensities are windowed between the 1st and 99.5th percentiles, which are estimated on every fourth pixel in each direction. Series are assigned to six slots by plane × fat suppression. Building the cache of all training series produced no decoding errors; series with fewer than three slices are excluded.

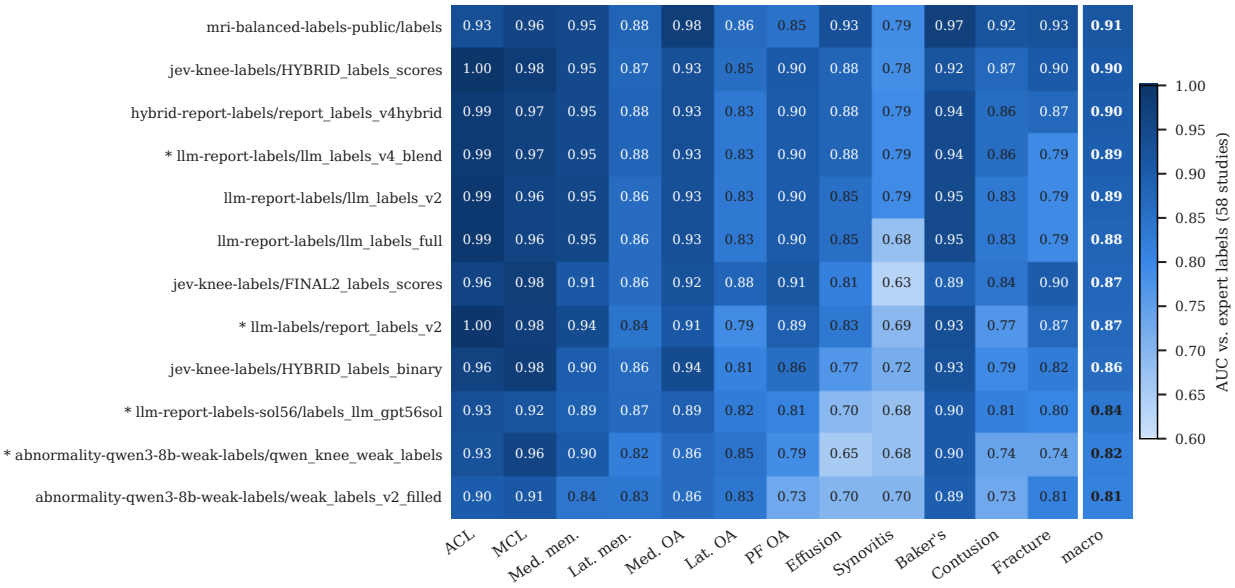


FIGURE 6. **Per-finding ROC AUC of public report-label tables against the expert labels of the 58 reference studies, sorted by macro AUC (right column). ∗: the four tables averaged into the reader's targets v0. Three tables flagged as potentially reference-contaminated are excluded; duplicate tables are shown once. Descriptive; $n = 58$.**

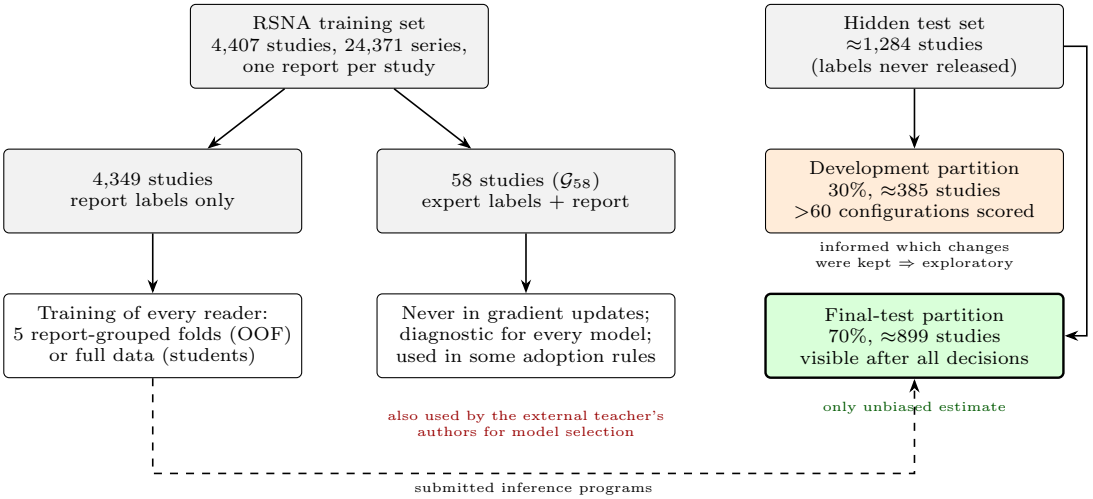


FIGURE 7. **Use of the data. Left: the training set; every reader is trained on the 4,349 report-labelled studies, and the 58 expert-labelled studies serve as a development diagnostic. Right: the hidden test set; the development partition informed model development, and the final-test partition was seen only after all decisions.**

C. 2.5D MULTI-SLOT READER (REUSED)

From each available slot s, K window centres are spread over 4–96% of the series (stratified-random in training, uniform at evaluation); each centre j gives a three-slice window $\mathbf{x}_{s,k} = [v_{j-1}, v_j, v_{j+1}]$ encoded by an ImageNet-pre-trained ConvNeXt-T or ConvNeXt-S [41], [42]. Each feature is projected to $d = 384$ and tagged with a slot embedding and a position embedding: $\mathbf{h}_{s,k} = \mathrm{LN}(Wf(\mathbf{x}_{s,k})) + \mathbf{e}_s^{\mathrm{slot}} + g(p_{s,k})$. All $6K$ tokens pass through a two-layer transformer encoder [43], and each finding has its own attention pooling [44]:

$$a_{c,n} = \underset{n}{\mathrm{softmax}}\left(\mathbf{u}_c^\top \tanh(V\mathbf{h}_n)\right), \tag{2}$$

$$z_c = \mathbf{w}_c^\top \textstyle\sum_n a_{c,n}\mathbf{h}_n + b_c. \tag{3}$$

Fig. 8 shows the architecture and Algorithm 3 the forward pass.

D. EXECUTED TRAINING CONFIGURATION

Every reader reported in Table 7 was trained by the released scripts listed in the Data and Code Availability section with the configuration of Table 4. These scripts differ from the upstream reference trainer in two respects: fp16 with loss scaling

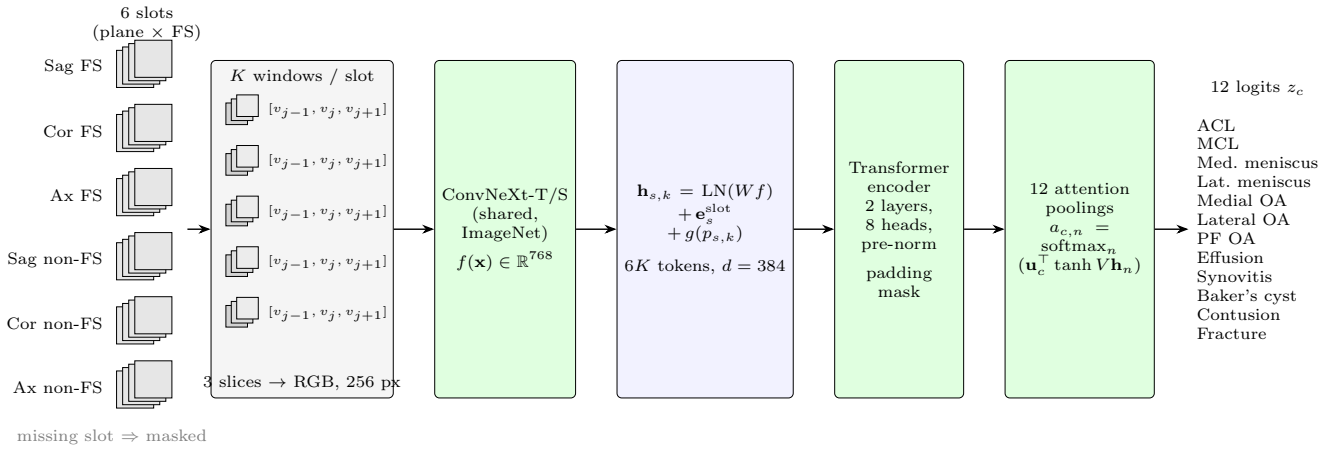


FIGURE 8. **The reused 2.5D multi-slot reader [9]. Each available plane/fat-suppression slot contributes K three-slice windows; a shared ConvNeXt encodes each window, slot and slice-position embeddings are added, a two-layer transformer mixes all tokens of the study, and one attention pooling per finding produces 12 logits. Missing slots are masked.**

Algorithm 2 Canonical volume for one DICOM series (reused)

Input: slices $\{(I_j, \mathbf{o}_j, \mathbf{p}_j, \boldsymbol{\delta}_j)\}$ (pixels, orientation, position, spacing)
Output: uint8 volume $V \in \{0..255\}^{n\times 384\times 384}$, $n \le 64$

invert MONOCHROME1; apply rescale slope and intercept
$\mathbf{n} \leftarrow \mathbf{o}^{\text{row}} \times \mathbf{o}^{\text{col}}$; plane $\leftarrow \arg\max_a |n_a|$
sort slices by $\sigma\,\langle \mathbf{p}_j, \mathbf{n}\rangle$ ▷ σ: canonical direction
if $n > 64$ **then** keep 64 uniformly spaced slices
end if
for each slice j **do**
    transpose/flip I_j to the nearest canonical LPS axes
    crop the central 153.6 mm; resample to 0.4 mm; zero-pad to 384^2
end for
$(\ell, h) \leftarrow$ percentiles $(1, 99.5)$ of $V[:, ::4, ::4]$
return $\text{round}\left(255 \cdot \text{clip}((V-\ell)/(h-\ell), 0, 1)\right)$

Algorithm 3 Forward pass of the 2.5D multi-slot reader (reused)

Input: canonical volumes $\{V_s\}$ of one study; windows per slot K; training flag
Output: logits $\mathbf{z} \in \mathbb{R}^{12}$

$T \leftarrow [\,]$
for each slot $s = 1, \ldots, 6$ with a series **do**
    $\{j_k\}_{k=1}^{K} \leftarrow \text{WindowCentres}(|V_s|, K, \text{training})$
    for $k = 1, \ldots, K$ **do**
        $\mathbf{x} \leftarrow \text{View}([V_s[j_k-1], V_s[j_k], V_s[j_k+1]])$ ▷ crop, resize 256; jitter if training
        append $\text{LN}(Wf(\mathbf{x})) + \mathbf{e}_s^{\text{slot}} + g(j_k/(|V_s|-1))$ to T
    end for
end for
$H \leftarrow \text{Transformer}_2(T)$ ▷ padding mask for missing slots
for $c = 1, \ldots, 12$ **do**
    $\mathbf{a}_c \leftarrow \text{softmax}(\mathbf{u}_c^\top \tanh(VH^\top))$; $z_c \leftarrow \mathbf{w}_c^\top (H^\top \mathbf{a}_c) + b_c$
end for
return $\mathbf{z}$

and gradient accumulation (the upstream trainer uses bf16 and direct batches, which the T4 does not support efficiently). The kept checkpoint is the exponential moving average after the last epoch, so no validation data select it. The EMA decay at update t is $\min\left(0.998, (1+t)/(10+t)\right)$, a warm-up that reaches 0.998 after about 4,000 updates. Augmentation is applied per series: rotation $\pm 12°$, scale $\pm 12\%$, translation $\pm 6\%$, gamma and gain jitter.

E. DISTILLATION TARGETS AND TEACHER PROVENANCE

A student is trained on all 4,349 studies outside $\mathcal{G}_{58}$ with targets

$$\tilde{y}_{ic} = \text{clip}\left((1-\tau)\, y_{ic}^{\text{rep}} + \tau\, t_{ic},\ 0.01,\ 0.99\right), \qquad (4)$$

where t_{ic} is a teacher probability and τ the teacher weight ($\tau = 0$: report-only). A study the teacher did not score keeps y^{rep}.

Table 5 records what is known about each teacher. We call the external teachers *image-input* teachers: they read images, but most of their supervision is itself report-derived.

Because the external teachers were applied in-sample, their targets are not cross-fitted, and the student learns the teacher's ranking of studies the teacher was fitted to. This does not leak hidden-test information, but it rules out claims of fully cross-fitted supervision. Algorithm 4 gives the procedure.

F. RANK-LEVEL FUSION

Because the metric depends only on within-finding ranks, students are fused with an existing model or ensemble P by normalised ranks. With $r(\cdot) \in [0, 1]$ the normalised rank over

TABLE 4. Executed training configuration of the 2.5D reader.

Setting	Value
Backbone	ConvNeXt-T / ConvNeXt-S, ImageNet
Parameters	28M / 50M backbone + ≈3M head
Input	6 slots × K windows, 3 slices, 256 px
Windows K	8 (training), 12 (evaluation)
Head	$d = 384$, 2 transformer layers, 8 heads
Optimiser	AdamW [45], weight decay 0.02
Learning rate	10^{-4} backbone, 5×10^{-4} head
Schedule	1 warm-up epoch, cosine decay, 14 epochs
Batch	8 studies = micro-batch 2 × 4 accumulation
Regularisation	drop-path 0.1, dropout 0.2, grad clip 3
Precision	fp16 with loss scaling
Weight averaging	EMA, decay $\min(0.998, \frac{1+t}{10+t})$
Hardware / time	1 NVIDIA T4 per run, 5.6–7.9 h

Algorithm 4 Teacher–student distillation

Input: targets y^{rep}, folds ϕ (Alg. 1); teacher type; weight τ; seed
Output: student parameters θ_S

if teacher is self **then**
  for $k = 0, \ldots, 4$ **do**
    train reader θ_k on $\{i : i \notin \mathcal{G}_{58},\ \phi(i) \neq k\}$ with targets y^{rep}
    $t_i \leftarrow \sigma(f_{\theta_k}(X_i))$ for all i with $\phi(i) = k$ ▷ out-of-fold
  end for
else
  $t_i \leftarrow \text{Teacher}(X_i)$ for every $i \notin \mathcal{G}_{58}$ ▷ external model, in-sample
end if
for each $i \notin \mathcal{G}_{58}$ **do**
  $\tilde{y}_i \leftarrow \text{clip}((1-\tau)y_i^{\text{rep}} + \tau t_i, 0.01, 0.99)$ if t_i defined, else y_i^{rep}
end for
initialise θ from ImageNet; $\bar{\theta} \leftarrow \theta$; $t \leftarrow 0$
for epoch $= 1, \ldots, 14$ **do**
  for each batch $B \subset \{i \notin \mathcal{G}_{58}\}$ **do**
    $\mathcal{L} \leftarrow \frac{1}{|B|} \sum_{i \in B} \text{BCE}(f_\theta(X_i), \tilde{y}_i)$ ▷ Alg. 3
    AdamW step on θ; $t \leftarrow t + 1$; $d \leftarrow \min(0.998, \frac{1+t}{10+t})$; $\bar{\theta} \leftarrow d\,\bar{\theta} + (1-d)\,\theta$
  end for
end for
return $\theta_S \leftarrow \bar{\theta}$

the test studies and $\bar{S}_{ic}$ the probability mean of M students,

$$F_{ic} = (1-w)\, r\big(P_{\cdot c}\big)_i + w\, r\big(\bar{S}_{\cdot c}\big)_i. \tag{5}$$

$w = 0.2$ was specified before the first fusion experiment, following a public notebook. Later, $w = 0.3$ and 0.4 were evaluated on the development partition (four submissions on two parents); none ranked above the corresponding $w = 0.2$ system, and $w = 0.2$ was kept for every subsequent system. The inference safeguards (checkpoint hashes, time guard, fallback to the parent's predictions, rank imputation for unscored studies) are given in Supplementary Algorithm S1. Fused scores are ranks over a cohort, not calibrated probabilities (Section VII).

V. EXPERIMENTAL SETUP

Implementation

PyTorch and timm [42]; training on single NVIDIA T4 GPUs (16 GB), two arms in parallel on a two-GPU machine; a DICOM cache of all training series built in five CPU shards (53.6 GB).

Protocol

Each experiment changed one variable relative to a parent with a known development score. Its hypothesis and adoption rule were recorded in a version-controlled experiment log before the submission was scored (for example, "adopt if the stand-alone development AUC is at least the parent's $+0.003$ and $\mathcal{G}_{58}$ does not fall by more than 0.01"). The log was not registered externally, so we describe it as prospectively logged rather than pre-registered. It is released with the code [39].

Baselines

Eleven published methods for the same task whose inference code is public and whose development score is displayed with the code [11]–[13], [38], [47]–[51]. We report their scores as published and did not re-train them.

Statistics

$\mathcal{G}_{58}$ macro AUCs use 2,000 study-level bootstrap resamples [52]; a resample in which a finding lacks positives or negatives contributes no AUC for that finding, and the macro AUC averages the remaining findings. Patient identifiers are unavailable, so patient-level resampling was not possible. Paired differences are computed on the same resamples. We do not interpret overlapping intervals or equal rounded scores as equivalence.

VI. RESULTS

A. FINAL-TEST PARTITION

Table 6 lists the comparisons that the final-test partition can answer. All model and selection decisions were made before any final-test score was visible. The rows are grouped so that each comparison changes one thing: stand-alone readers that differ only in the training target, and *matched* ensemble pairs in which the same parent was submitted with and without students. The two submitted final systems (the unchanged parent C2 and the fused system H976, which adds students *and* other switches) differ in several components, so their comparison is reported as a system-level result and is not used as evidence for the students. H976 was the best of 23 configurations displayed at 0.958, so its development score is optimistically biased.

[TODO: Fill the final-test column from the Kaggle submission list (field privateScore) and describe the four matched pairs and the A/B/C/C3 comparison here. If a row has no

TABLE 5. Teacher provenance. "In-sample": the teacher had been trained on the studies it labelled for the student. Entries marked *unverified* are as described by the component's authors and could not be checked against code or data.

Teacher	Training images	Target labels	External data / pre-training	$\mathcal{G}_{58}$ use	Predictions used
Self (arm B)	4,349 RSNA studies, 5-fold CV	report consensus (Eq. 1)	ImageNet only	diagnostic only; 3 of 5 fold models are public releases [9]	out-of-fold
Mirror (C, C2, C3, S75, T75s1)	4,349 RSNA studies + 2,399 OAI knees [11]	RSNA: blend of report-derived label sets; OAI: expert MOAKS grades for PF OA, lateral OA, synovitis only (other cells masked)	OAI [16]; ImageNet-pre-trained CoAtNet	excluded from training, used for selection; inference recipe tuned with development scores	in-sample
S1 component R6 (E75t, E75s)	RSNA training studies [12]	*unverified*	fastMRI auxiliary data [17] (*unverified*)	*unverified*	reported as out-of-fold (*unverified*)
S1 component S6fall-B	all RSNA training studies [46]	*unverified*	SKM-TEA anatomy model [18]	*unverified*	in-sample (all-data model)

TABLE 6. Development (30%) and final-test (70%) macro AUC. The final-test partition was not used for development or selection. ID: submission identifier.

System	ID	Dev.	Final test
Stand-alone readers (target is the only change)			
A: report targets	57053758	0.933	[TODO:]
B: self-distilled	57053761	0.939	[TODO:]
C: image-input teacher, $\tau = 0.5$	57081442	0.948	[TODO:]
C3: image-input teacher, $\tau = 0.75$	57103459	0.949	[TODO:]
Teacher alone [11]	–	0.949	[TODO:]
Matched pairs (students are the only change)			
P0: public CoAtNet + R6 blend	57252502	0.958	[TODO:]
P0 + 5 Mirror students	57281541	0.958	[TODO:]
P1 = C2 (P0 + S6fall-B)	57315983	0.958	[TODO:]
P1 + 5 Mirror students	57369206	0.958	[TODO:]
P1 + V2 weights	57369271	0.958	[TODO:]
+ 5 Mirror students	57369477	0.958	[TODO:]
P1 + V2 + BK + 5 students	57447421	0.958	[TODO:]
+ 2 S1 students	57497105	0.958	[TODO:]
Submitted final systems (several changes; system level only)			
C2 (unchanged parent P1)	57315983	0.958	[TODO:]
H976 (P1 + V2 + BK + 7 students)	57497105	0.958	[TODO:]

final-test score, say so; the corresponding development comparison then remains exploratory. If a paired-difference figure is added, plot final-test minus development differences per pair.]

B. TRAINING TARGETS (DEVELOPMENT PARTITION)

Table 7 and Fig. 9 list every full-data reader; each differs from its parent in one variable.

Observed

Self-distillation (B vs A) raised the development AUC from 0.933 to 0.939; replacing the self-teacher by the image-input teacher (C vs B) raised it to 0.948, and $\tau = 0.75$ (C3) displayed 0.949, the same value as the teacher at three-decimal precision. On $\mathcal{G}_{58}$, B differed from A by -0.003, and C from B by $+0.019$, with the largest per-finding changes on synovitis (0.717→0.791) and fracture (0.844→0.933). No clear benefit was observed from ImageNet-22k pre-training (B2 vs B, -0.001), from ConvNeXt-S (S75 vs C3 on $\mathcal{G}_{58}$; also -0.003 and -0.004 OOF in an earlier cross-validated comparison with report targets), or from a second seed (C2 vs C, equal at three decimals). Only two seeds were run, so seed variability is not characterised.

TABLE 7. Full-data 2.5D readers. τ: teacher weight in (4). $\mathcal{G}_{58}$: descriptive (Section III-B). Dev.: stand-alone development-partition AUC; "–" not evaluated alone.

Arm	Backbone	Target	τ	$\mathcal{G}_{58}$	Dev.
A	ConvNeXt-T	reports	0	0.907	0.933
B	ConvNeXt-T	+ self OOF	0.5	0.904	0.939
B2	ConvNeXt-T in22k	+ self OOF	0.5	0.907	0.938
C	ConvNeXt-T	+ Mirror	0.5	0.923	0.948
C2	ConvNeXt-T (seed 1)	+ Mirror	0.5	0.919	0.948
C3	ConvNeXt-T	+ Mirror	0.75	0.927	0.949
S75	ConvNeXt-S	+ Mirror	0.75	0.921	–
T75s1	ConvNeXt-T (seed 1)	+ Mirror	0.75	0.931	–
E75t	ConvNeXt-T	+ S1 ens.	0.75	0.926	–
E75s	ConvNeXt-S	+ S1 ens.	0.75	0.924	–
Mean of 5 Mirror students				0.926	–
Mean of 5 Mirror + 2 S1 students				0.928	–

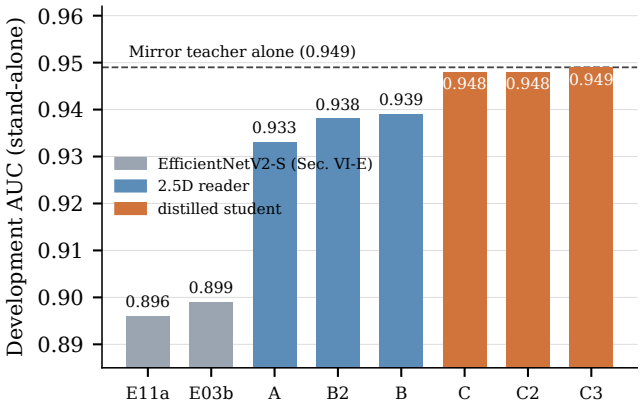


FIGURE 9. Stand-alone development-partition AUC of each model. E11a and E03b are the EfficientNetV2-S models of Section VI-E. The dashed line is the image-input teacher's displayed score.

TABLE 8. Development-partition macro AUC of published methods (as displayed with their code) and of this work. †: uses weights trained with external data (OAI, fastMRI or SKM-TEA). Descriptive only.

Method	Dev. AUC
Published single models	
CoAtNet MIL, report labels [38]	0.924–0.932
MaViT [47]	0.935
Anatomical-mirror CoAtNet MIL† [11]	0.949
Published ensembles	
DINOv2/DINOv3/RadImageNet/CoAtNet blend [48]	0.941
Re-weighted blend + extra readers [49]	0.943
Stack + 2.5D ConvNeXt MIL [13]	0.944
+ private EffNetV2-S/CoAtNet/ConvNeXt-L leg [50]	0.946
Mirror + 2.5D reader stack† [51]	0.950
Mirror + compact ConvNeXt (S6)† [12]	0.954
Mirror + compact ConvNeXt (R7)† [12]	0.956
CoAtNet 384/224 + R6 blend† [12], [13]	0.958
This work	
Reader, report targets (A)	0.933
Reader, self-distilled (B)	0.939
Reader, image-input teacher (C3)†	0.949
Final fused system†	0.958

TABLE 9. Students as the single change to an existing ensemble (Eq. (5), $w = 0.2$), development partition. "$>$": ranked above the parent in the unrounded ordering (ordinal, exploratory).

Parent ensemble	Added	Parent	Fused
Mirror + S6 compact	C	0.954	0.955
Mirror + R7 compact	C, C2, C3	0.956	0.957
Mirror + R7 compact	5 Mirror st.	0.956	0.957
CoAtNet + R6 blend	5 Mirror st.	0.958	0.958 (>)
+ S6fall-B (P1)	5 Mirror st.	0.958	0.958 (>)
P1 + V2 weights	5 Mirror st.	0.958	0.958 (>)
P1+V2+BK + 5 st.	2 S1 st.	0.958	0.958 (>)
P1+BK+FK + 5 st.	2 S1 st.	0.958	0.958 (>)

TABLE 10. EfficientNetV2-S@384 models (two folds). OOF: pooled over 1,740 studies against the soft training labels, with labels in the uncertain band 0.25–0.7 excluded († consensus labels; 0.873 against the labels of E11).

ID	Change	OOF	$\mathcal{G}_{58}$	Dev.
–	published ensemble (parent)	–	–	0.943
E11	parent + model, $w = 0.30$	0.872	0.900	0.935
E11a	model alone	0.872	0.900	0.896
E03b	alone, consensus labels	0.911†	0.883	0.899
E13	ResNet-34, own cache	0.843	0.865	–

Caveats

These are development-partition comparisons and, until confirmed on the final-test partition, are exploratory. The $\mathcal{G}_{58}$ changes of the Mirror students are not independent, because the teacher's authors selected it on $\mathcal{G}_{58}$. Three explanations for the C-vs-B gain are confounded and are not separated by our experiments: image evidence learned by the teacher, external data (OAI with expert grades for three findings), and complementary errors between teacher and reader. A teacher-only target ($\tau = 1$) and a teacher matched in architecture but without external data were not run.

Cost

We make no claim about inference cost: measured runtimes are not available for either the teacher or the students. (The teacher's authors report submit-to-score turnaround, which includes queueing and is explicitly not an inference runtime [11].) [TODO: Add measured per-study decode time, model time, peak memory and fallback counts for a single student and for the seven-student stage, from the inference logs.]

C. COMPARISON WITH PUBLISHED METHODS (DEVELOPMENT PARTITION)

Table 8 places the readers among published methods evaluated on the same development partition. A single distilled reader displayed the same AUC as the best published single model, the teacher it was distilled from. Because published scores were also obtained with development feedback, the table is descriptive.

D. FUSION WITH EXISTING ENSEMBLES (DEVELOPMENT PARTITION)

Table 9 lists every comparison in which the students were the only change to an existing ensemble. Three comparisons changed the displayed score (0.954→0.955 once, 0.956→0.957 twice, with three and with five students on the same parent); in the remaining five the displayed score was unchanged and the unrounded ordering placed the fused variant above its parent. These are eight ordinal, non-independent comparisons on the same partition—they share parents and students—so we report them as consistent with a small benefit that requires final-test confirmation, not as an established improvement. Further single-change switches inside the ensembles are reported in Supplementary Table S1.

E. NEGATIVE RESULTS: EFFICIENTNETV2-S ON A PUBLISHED CORPUS

Before the reader, we trained EfficientNetV2-S [53] at 384 px with stochastic weight averaging [54] on a published 44-slice corpus and its labels [38] (E11). Two folds reached an OOF AUC of 0.872 (soft-label estimand, Table 10) and $\mathcal{G}_{58}$ 0.900, but the model alone displayed 0.896, and adding it at a prospectively logged weight of 0.30 lowered a 0.943 ensemble to 0.935. A test-time preprocessing mismatch was excluded by rebuilding 218 training studies from DICOM: pixels matched, prediction ranks agreed on all 12 findings, and $\mathcal{G}_{58}$ AUC was identical. A four-source consensus target (E03b) displayed 0.899, and a ResNet-34 multi-view model on our own cache (E13) had a lower OOF AUC (0.843). Better OOF did not imply a useful ensemble member: the model shared data, labels and head design with the ensemble's dominant component.

TABLE 11. Per-finding OOF AUC of E11 against report labels binarised at 0.5 (folds 0+1, $n = 1{,}740$, no band excluded; macro 0.870) and positives in $\mathcal{G}_{58}$.

Finding	OOF	n_+	Finding	OOF	n_+
ACL	0.890	24	PF OA	0.845	21
MCL	0.846	9	Effusion	0.908	35
Med. meniscus	0.894	26	Synovitis	0.891	27
Lat. meniscus	0.816	23	Baker's cyst	0.926	12
Medial OA	0.885	15	Contusion	0.813	19
Lateral OA	0.832	11	Fracture	0.896	18

Table 11 gives E11's per-finding OOF AUC under a different estimand (labels binarised at 0.5, no band excluded; macro 0.870). Lateral meniscus, contusion, lateral OA and PF OA had the lowest agreement with report labels.

F. RESOLUTION OF THE DEVELOPMENT YARDSTICKS

$\mathcal{G}_{58}$

For E11, the $\mathcal{G}_{58}$ macro AUC of 0.900 has a 95% bootstrap interval of [0.872, 0.924]; for E03b, 0.883 has [0.846, 0.915]; their paired difference of -0.017 has $[-0.042, +0.003]$. These intervals describe E11 and E03b, whose $\mathcal{G}_{58}$ predictions are stored; the predictions of A, B and C3 on $\mathcal{G}_{58}$ were not retained locally, so paired intervals for the principal comparisons are not reported. [TODO: Required before submission: download `gold_arm{A,B,C,C3}.csv` from the training-kernel outputs and report the paired intervals for B−A, C−B and C3−A computed by the released script; they remain descriptive because $\mathcal{G}_{58}$ was used in development.]

Development partition (illustration)

To illustrate how much a 385-study partition can resolve, we used E11's OOF predictions (1,740 studies, report labels binarised at 0.5) as a stand-in population. Under these assumptions, the macro AUC of a fixed model has a standard deviation of 0.0085 across random 385-study subsets (Fig. 10). We then created a second model by adding Gaussian noise to E11's predictions, so that its full-sample AUC was lower by a known gap, and counted how often a subset ordered the two wrongly: gaps of 0.0003, 0.0012 and 0.0037 were misordered 25%, 10% and 2% of the time. These rates depend on the chosen model, perturbation, prevalence, correlation between models and label noise; they illustrate, rather than estimate, the resolution of the hidden partition. Repeated comparisons on the same partition are not independent replications.

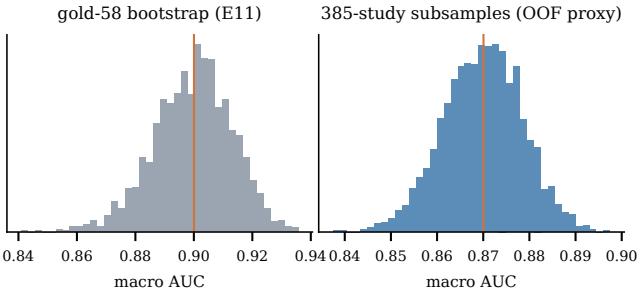


FIGURE 10. Illustration of yardstick resolution. Left: bootstrap distribution of E11's $\mathcal{G}_{58}$ macro AUC. Right: macro AUC of the same model on random 385-study subsets of its OOF predictions (report labels). Vertical lines: full-sample values.

VII. DISCUSSION

Findings

On the development partition, the training target was associated with larger differences than the backbone, pre-training or seed in the configurations we evaluated; distillation from an image-input teacher gave the largest change, and fused students ranked above their parent ensembles in every development comparison. Whether these differences hold on the final-test partition is reported in Section VI-A.

Possible explanations (not tested)

The teacher may encode image evidence that reports omit; it was also trained with external data that include expert grades for three findings, and teacher and reader may make complementary errors. Our experiments do not separate these mechanisms (Section VI-B). The development gain of fused students may reflect the different representation of the reader or the report term in (4), which keeps the student partly anchored to reports.

Relation to prior work

Kooklan [30] found that a report–image mixture agreed better with the expert labels than either source, but did not improve a 2.5D ResNet-50 on the final-test partition. Our development gains point the other way; the final-test scores of our readers are needed before concluding that the difference reflects the reader, the teacher, or development-partition over-fitting.

Clinical considerations

A macro AUC around 0.95 on the development partition does not establish clinical usefulness. Clinical prioritisation is a potential application that would require per-finding sensitivity and specificity at thresholds fixed on development data, precision–recall analysis for uncommon findings, calibration, review of false positives and negatives by a qualified reader, and subgroup and scanner analyses on an independent cohort with compatible label definitions. Rank-fused scores are not probabilities, and their values depend on the cohort being ranked; a single newly arriving examination would require either a reference cohort against which to rank it or the students' own (uncalibrated) probabilities.

Ethical and data considerations

The dataset is de-identified and was used under its terms of use. [TODO: State whether your institution required ethics review or granted an exemption.] Components marked † were trained by their authors with external research data (OAI, fastMRI or SKM-TEA; Table 5); the dataset's rules allowed external data only under conditions that each participant had to verify. Our results without such weights are the report-target reader A (0.933) and the self-distilled reader B (0.939).

VIII. LIMITATIONS

(i) Except Table 6, all scores come from a 385-study development partition that informed development; they are exploratory. (ii) $\mathcal{G}_{58}$ is small and enriched, was used in adoption rules, and was used by the teacher's authors for selection. (iii) External teachers were applied in-sample, and their provenance is partly unverified (Table 5). (iv) Ablations needed to explain the main gain were not run: teacher-only targets ($\tau = 1$), a teacher without external data, at least three seeds of A, B and C3, and individual label sources versus their consensus. (v) Institutions, acquisition period, demographics, scanners, inclusion criteria, patient identifiers and the selection of $\mathcal{G}_{58}$ were not reported by the organisers (label definitions and the reading procedure were; Table 1). (vi) No external testing, calibration or reader study was performed. (vii) The fused system depends on published models whose training we could only partly audit.

IX. CONCLUSION

For a report-supervised 2.5D knee MRI reader with 58 expert-labelled studies, changing the training target from report consensus to a mixture with an image-input teacher was associated with a development-partition AUC increase from 0.933 to 0.949, while the backbone, pre-training and seed changes we evaluated showed no clear benefit; students fused with existing ensembles ranked above their parents in all development comparisons. These findings are exploratory until confirmed on the final-test partition (Section VI-A) and on external data, and the mechanism of the gain remains to be isolated with teacher-only and external-data-matched controls.

DATA AND CODE AVAILABILITY

The dataset is available from the RSNA under its terms of use [8]. Code, the executed training scripts of all readers except the two S1 students ([TODO: add the script of Kaggle kernel `hosen42/rsna-knee-gpjw-student5`]), inference scripts, fold assignments, the experiment log and the scripts that regenerate all figures are at https://github.com/hamidhosen42/RSNA-Knee-Abnormality-Detection (folder `ieee_access_paper`, [TODO: tag a release and cite its version/DOI]). [TODO: State where trained checkpoints (with SHA-256) can be obtained.] Published components remain under their authors' licences.

USE OF AI TOOLS

[TODO: Confirm and adapt:] Portions of the manuscript text, figures and analysis code were drafted with the assistance of an AI system (Claude, Anthropic). The authors reviewed and edited all content and take full responsibility for it.

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